Theory Of Computation-CSA1303

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GitHub Link:-https://github.com/SUMANTH111111/CSA1303

Experiment -2:-

Code:-

```
#include <stdio.h>
#include <string.h>
int main() {
    char str[100];
    int state = 0;
    printf("Enter a string over {0, 1}: ");
    scanf("%s", str);
    for (int i = 0; str[i] != '\0'; i++) {
        switch (state) {
            case 0:
                if (str[i] == '0')
                    state = 1;
                else if (str[i] == '1')
                    state = 3;
                else {
                    printf("Invalid input! Use only 0 and 1.\n");
                    return 0;
                }
                break;
            case 1:
                if (str[i] == '0')
                    state = 1;
                else
                    state = 2;
                break;
            case 2:
                if (str[i] == '0')
```

```
                state = 1;
            else
                state = 2;
            break;

        case 3:
            state = 3;
            break;
    }
  }
  if (state == 2)
    printf("String Accepted\n");
  else
    printf("String Rejected\n");
  return 0;
}
```

OUTPUT:-

